

Prior Work: summarized references

Summarized from Zotero Report Prior Work.htm. Abstracts were replaced with plain-language summaries that emphasize findings, numbers, and conclusions where available. Fields such as Item Type, Library Catalog, URL, Accessed, Extra, ISSN, Date Added, Modified, and Tags were omitted.

1. IMPACT BACK STUDY PROTOCOL. IMPLEMENTATION OF SUBGROUPING FOR TARGETED TREATMENT SYSTEMS FOR LBP PATIENTS IN PRIMARY CARE: A PROSPECTIVE POPULATION-BASED SEQUENTIAL COMPARISON

First Author: Nadine E. Foster

Summary:

This publication is the trial protocol only: it states planned recruitment, outcomes, and analysis for IMPaCT Back but does not report patient results inside this item.

Tool: IMPaCT

Date: 2010-08-20 Publication: BMC Musculoskeletal Disorders Volume: 11 Issue: 1 Pages: 186
DOI: 10.1186/1471-2474-11-186 <http://doi.org/10.1186/1471-2474-11-186>

2. COMPARISON OF STRATIFIED PRIMARY CARE MANAGEMENT FOR LBP WITH CURRENT BEST PRACTICE (START BACK): A RANDOMISED CONTROLLED TRIAL

First Author: Jonathan C. Hill

Summary:

At four months, the stratified-care group improved more on disability scores compared to usual care (adjusted mean difference about 1.8 points on the Roland-Morris scale), and a smaller but still significant gap remained at twelve months (about 1.1 points), matching small-to-moderate effect sizes. By twelve months the intervention also yielded roughly 0.04 extra quality-adjusted life years per person and lower back-pain-related costs in the analysis shown.

Tool: STaRT

Date: 2011-10-29 Publication: Lancet Volume: 378 Issue: 9802 Pages: 1560-1571 Place: London, England DOI: 10.1016/S0140-6736(11)60937-9 [http://doi.org/10.1016/S0140-6736\(11\)60937-9](http://doi.org/10.1016/S0140-6736(11)60937-9)

3. EFFECT OF STRATIFIED CARE FOR LBP IN FAMILY PRACTICE (IMPACT BACK): A PROSPECTIVE POPULATION-BASED SEQUENTIAL COMPARISON

First Author: Nadine E. Foster

Summary:

Stratified care produced a statistically significant but small average disability improvement versus usual care (mean Roland-Morris difference about 0.7 points), with a much larger effect in the high-risk subgroup (about 2.3 points). Workers took about half as many sick days (median 4 vs 8), sick notes dropped (9% vs 15%), and the economic analysis suggested cost savings alongside small utility gains.

Tool: IMPaCT

Date: 2014/03/01 Publication: The Annals of Family Medicine Volume: 12 Issue: 2 Pages: 102–111
DOI: 10.1370/afm.1625 <http://doi.org/10.1370/afm.1625>

4. ESTIMATING THE RISK OF CHRONIC PAIN: DEVELOPMENT AND VALIDATION OF A PROGNOSTIC MODEL (PICKUP) FOR PATIENTS WITH ACUTE LBP

First Author: Adrian C. Traeger

Summary:

In external validation data the PICKUP model discriminated people who developed chronic pain from those who did not with moderate accuracy (AUC about 0.66 with a 95% confidence interval excluding 0.5). Decision-curve analysis suggested risk-based screening could avoid many unnecessary interventions compared with treating everyone the same, though calibration still drifted for some high-risk groups.

Tool: PICKUP

Date: May 17, 2016 Publication: PLOS Medicine Volume: 13 Issue: 5 Pages: e1002019 DOI: 10.1371/journal.pmed.1002019 <http://doi.org/10.1371/journal.pmed.1002019>

5. A DATA-DRIVEN PREDICTIVE SYSTEM USING CASE-BASED REASONING FOR THE CONFIGURATION OF DEVICE-ASSISTED BACK PAIN THERAPY

First Author: Juan A. Recio-García

Summary:

Compared with a standard interpolation setup, the case-based reasoning engine raised successful machine configuration rates from about 31% to about 70% in the authors' tests. In a longitudinal observational run, patients averaged about 32% improvement on a pain scale, about 7% on the Oswestry disability index, and about 13% on a quality-of-life measure.

Tool: Case-Based Reasoning (CBR) software

Date: 2021-07-04 Publication: Journal of Experimental & Theoretical Artificial Intelligence
Volume: 33 Issue: 4 Pages: 617–635 DOI: 10.1080/0952813X.2019.1704441
<http://doi.org/10.1080/0952813X.2019.1704441>

6. EFFECTIVENESS OF APP-DELIVERED, TAILORED SELF-MANAGEMENT SUPPORT FOR ADULTS WITH LOWER BACK PAIN-RELATED DISABILITY: A SELFBACK RANDOMIZED CLINICAL TRIAL

First Author: Louise Fleng Sandal

Summary:

In 461 adults analyzed, the selfBACK app group scored about 0.8 points lower on the Roland-Morris disability scale at three months than usual care alone (statistically significant but modest gap). About 52% of app users versus 39% of controls improved by at least four points; the authors flag that the clinical importance of the average difference is uncertain.

Tool: selfBACK

Date: 2021-10-01 Publication: JAMA Internal Medicine Volume: 181 Issue: 10 Pages: 1288-1296
DOI: 10.1001/jamainternmed.2021.4097 <http://doi.org/10.1001/jamainternmed.2021.4097>

7. USABILITY TESTING OF A DIGITAL ASSESSMENT ROUTING TOOL FOR MUSCULOSKELETAL DISORDERS: ITERATIVE, CONVERGENT MIXED METHODS STUDY

First Author: Cabella Lowe

Summary:

Twenty-two participants tested four DART iterations over five rounds (50 clinical presentations). Usability problems fell from 12 to zero; mean System Usability Scale score was 84.3 (excellent range). Mean completion time was 5.2 minutes with no DART system errors. Trust and intention to act improved during the study. The authors concluded serious usability barriers were resolved, supporting planned RCTs of safety and effectiveness versus usual care.

Tool: DART

Date: 2022-08-30 Publication: Journal of Medical Internet Research Volume: 24 Issue: 8 Pages: e38352 DOI: 10.2196/38352 <http://doi.org/10.2196/38352>

8. RISK-BASED STRATIFIED PRIMARY CARE FOR COMMON MUSCULOSKELETAL PAIN PRESENTATIONS (STarT MSK): A CLUSTER-RANDOMISED, CONTROLLED TRIAL

First Author: Jonathan C. Hill

Summary:

This UK cluster trial (24 practices) found no meaningful difference between STarT MSK-supported care and usual care on the primary pain and function outcomes averaged over six months (e.g. pain scores 4.4 vs 4.6). Process measures changed—intervention patients more often received information, physio referrals, and OTC analgesics—but overall pain and function did not improve significantly.

Tool: STarT

Date: 2022-09-01 Publication: The Lancet Rheumatology Volume: 4 Issue: 9 Pages: e591-e602 DOI: 10.1016/S2665-9913(22)00159-X [http://doi.org/10.1016/S2665-9913\(22\)00159-X](http://doi.org/10.1016/S2665-9913(22)00159-X)

9. DEVELOPMENT AND INTERNAL VALIDATION OF A MACHINE LEARNING PREDICTION MODEL FOR LBP NON-RECOVERY IN PATIENTS WITH AN ACUTE EPISODE CONSULTING A PHYSIOTHERAPIST IN PRIMARY CARE

First Author: J. Knoop

Summary:

Compared results with STarT Back tool and physiotherapists expectations. Among 247 patients with acute LBP, about 47% still had meaningful pain at three months. The best machine-learning model reached modest discrimination for non-recovery (AUC about 0.66), similar to a traditional logistic model but better than usual-practice benchmarks such as therapists' expectations and the STarT Back tool in this sample. External validation is still required.

Tool: unspecified– likely Decision Trees

Date: 2022-09-03 Publication: BMC musculoskeletal disorders Volume: 23 Issue: 1 Pages: 834
DOI: 10.1186/s12891-022-05718-7 <http://doi.org/10.1186/s12891-022-05718-7> PMID: 36057717
PMCID: PMC9440317

10. MACHINE LEARNING CLINICAL DECISION SUPPORT FOR INTERDISCIPLINARY MULTIMODAL CHRONIC MUSCULOSKELETAL PAIN TREATMENT

First Author: Fredrick Zmudzki

Summary:

Across thirteen outcomes, decision tree models achieved AUC values roughly between 0.49 and 0.65 under cross-validation. Combining all predictions into one patient-level profile still agreed with clinician-judged outcomes for about three-quarters of patients, and chart review of “predicted negative” cases supported the algorithms more often than chance.

Tool: Decision Trees

Date: 2023-05-09 Publication: Frontiers in Pain Research Volume: 4 DOI:
10.3389/fpain.2023.1177070 <http://doi.org/10.3389/fpain.2023.1177070>

11. EVALUATION OF AN ELECTRONIC CLINICAL DECISION SUPPORT SYSTEM (DESSBACK) TO IMPROVE LBP MANAGEMENT: A PILOT CLUSTER RANDOMIZED CONTROLLED TRIAL

First Author: Abdul Malik Tun Firzara

Summary:

Thirty-six primary-care patients (23 intervention, 13 control) completed two-month follow-up. Fidelity was weak on the patient side but strong among physicians. Disability and anxiety favored DeSSBack with medium effect sizes (about 0.72 and 0.48), while pain and depression changes were small. Interviews showed doctors felt the tool standardized care, matched treatment to risk, saved time, and was easy to use.

Tool: DeSSBack

Date: 2023-05-27 Publication: Family Practice Volume: 40 Issue: 5-6 Pages: 742-752 DOI:
<https://doi.org/10.1093/fampra/cmado44>

12. NATURAL LANGUAGE PROCESSING OF REFERRAL LETTERS FOR MACHINE LEARNING–BASED TRIAGING OF PATIENTS WITH LBP TO THE MOST APPROPRIATE INTERVENTION: RETROSPECTIVE STUDY

First Author: Sebastian Fudickar

Summary:

Adding NLP-extracted data from referral letters raised triage F1 scores by up to about 20 percentage points for some referral categories in 1,608 patients, yet overall triage accuracy remained modest and the authors still judged the models below the bar for standalone clinical use

Tool: SVM, K-nn, aNN

Date: 2024-01-30 Publication: Journal of Medical Internet Research Volume: 26 Issue: 1 Pages: e46857 DOI: 10.2196/46857 <http://doi.org/10.2196/46857>

13. DIAGNOSTIC ACCURACY OF A LARGE LANGUAGE MODEL IN RHEUMATOLOGY: COMPARISON OF PHYSICIAN AND CHATGPT-4

First Author: Martin Krusche

Summary:

On the Gräf et al. (2022) case set, ChatGPT-4 matched the top diagnosis as often as rheumatologists (35% vs 39%; not significant) and was similar for top-three diagnoses (60% vs 55%). In inflammatory rheumatic disease (IRD) cases, ChatGPT-4 led on top diagnosis (71% vs 62%) and top-three (86% vs 74%). In non-IRD cases, physicians had higher top-diagnosis accuracy (27% vs 15%). Considering only the first diagnostic suggestion, ChatGPT-4 classified 58% of cases as IRD vs 56% for rheumatologists (not significant). The pattern shows higher sensitivity for IRD with lower specificity in non-IRD cases, with ChatGPT matching or outperforming rheumatologists.

Tool: ChatGPT-4

Date: 2024-02-01 Publication: Rheumatology International Volume: 44 Issue: 2 Pages: 303-306 DOI: 10.1007/s00296-023-05464-6 <http://doi.org/10.1007/s00296-023-05464-6>

14. EFFECTIVENESS OF A CLASSIFICATION-BASED APPROACH TO LBP IN PRIMARY CARE: A BENCHMARKING CONTROLLED TRIAL

First Author: Anna Sofia SIMULA

Summary:

At twelve months the classification-based pathway was not superior to usual care on the main physical-function outcome. Pain intensity improved sooner in the intervention arm (a significant time-by-group pattern), and the intervention group used less healthcare, imaging, and sick leave over follow-up.

Tool: STarT

Date: 2024-4-20 Publication: Journal of Rehabilitation Medicine Volume: 56 Pages: 28321 DOI: 10.2340/jrm.v56.28321 <http://doi.org/10.2340/jrm.v56.28321> PMID: 38643363 PMCID: PMC11151494

15. DECISION SUPPORT FOR MANAGING COMMON MUSCULOSKELETAL PAIN DISORDERS: DEVELOPMENT OF A CASE-BASED REASONING APPLICATION

First Author: Fredrik Granviken

Summary:

The SupportPrim PT prototype could retrieve clinically similar prior patients from a Norwegian primary-care physiotherapy case base using weighted prognostic features. When comparing the similarity scores for all patients in the case base, one at a time, with the ÖMSPQ and MSK-HQ, the most similar patients had a mean absolute difference from the query patient of 9.3 (95% CI

8.0–10.6) points on the ÖMSPQ and a mean absolute difference of 5.6 (95% CI 4.6–6.6) points on the MSK-HQ

Tool: SupportPrim

Date: 2024-05-10 Publication: JMIR Formative Research Volume: 8 Issue: 1 Pages: e44805 DOI: 10.2196/44805 <http://doi.org/10.2196/44805>

16. EVALUATION OF A MUSCULOSKELETAL DIGITAL ASSESSMENT ROUTING TOOL (DART): CROSSOVER NONINFERIORITY RANDOMIZED PILOT TRIAL

First Author: Cabella Lowe

Summary:

Seventy-eight patients completed the pilot crossover: raw agreement between DART and physiotherapist triage was weak to moderate, but a service-adjusted analysis raised concordance to about 78% with no harmful triage episodes detected. Mean usability scores were in the “excellent” range.

Tool: DART

Date: 2024-7-30 Publication: JMIR Formative Research Volume: 8 Pages: e56715 DOI: 10.2196/56715 <http://doi.org/10.2196/56715> PMID: 39078682 PMCID: PMC11322692

17. IMPROVING MUSCULOSKELETAL CARE WITH AI ENHANCED TRIAGE THROUGH DATA DRIVEN SCREENING OF REFERRAL LETTERS

First Author: Tjardo Daniël Maarseveen

Summary:

On thousands of referral letters, models for rheumatoid arthritis reached AUC-ROC near 0.78, osteoarthritis near 0.71, fibromyalgia near 0.81, and chronic follow-up near 0.63 when validated across sites. The RA classifier prioritized true RA cases better than the manual triage process; authors argue the pipeline could reduce workload.

Tool: XGBoost decision trees

Date: 2025-02-14 Publication: npj Digital Medicine Volume: 8 Issue: 1 Pages: 98 DOI: 10.1038/s41746-025-01495-4 <http://doi.org/10.1038/s41746-025-01495-4> License: 2025 The Author(s)

18. IMPROVING SAFETY, EFFICIENCY, COST, AND SATISFACTION ACROSS A MUSCULOSKELETAL PATHWAY USING THE DIGITAL ASSESSMENT ROUTING TOOL FOR TRIAGE: QUALITY IMPROVEMENT STUDY

First Author: Cabella Lowe

Summary:

Quality-improvement evaluation of DART as first contact for NHS MSK self-referrals (4,076 patients; three study cycles, February–May 2024). Agreement between clinicians and DART was 96%; no safety incidents occurred; 9.8% of patients were flagged urgent; 203 fewer cases needed escalation after initial clinician contact versus prestudy patterns; administrative time to process

referrals fell 51%, with about £80 saved per 100 referrals (authors' estimate). Mean patient satisfaction was about 90%. Service staff rated the pathway as a positive change.

Tool: DART

Date: 2025-04-25 Publication: Journal of Medical Internet Research Volume: 27 Issue: 1 Pages: e67269 DOI: 10.2196/67269 <http://doi.org/10.2196/67269>

19. PERSONALISED DECISION SUPPORT IN THE MANAGEMENT OF PATIENTS WITH MUSCULOSKELETAL PAIN IN PRIMARY PHYSIOTHERAPY CARE: A CLUSTER RANDOMISED CONTROLLED TRIAL (THE SUPPORTPRIM PROJECT)

First Author: Fredrik Granviken

Summary:

At twelve weeks, roughly 55% of both arms reported feeling better on the global improvement scale (odds ratio near 1.2, not significant), while the decision-support group actually did worse on one function scale than usual care. The trial therefore did not show benefit for the AI-supported pathway on the primary end points.

Tool: SupportPrim

Date: May 2025 Publication: PAIN Volume: 166 Issue: 5 Pages: 1167 DOI: 10.1097/j.pain.0000000000003456 <http://doi.org/10.1097/j.pain.0000000000003456>

OVERALL SYNTHESIS (FROM SUMMARIES, EMPHASIZING BOLDED CONCLUSIONS)

i) Main findings

Stratified primary care for LBP has mixed trial evidence: classic STarT Back and IMPaCT reports describe greater disability improvement than usual care (small to moderate effects, with larger benefits in higher-risk subgroups in IMPaCT) and economic or work-related advantages in some analyses. Later or different implementations do not always replicate: STarT MSK showed no meaningful difference on primary pain and function outcomes; a classification-based pathway was not superior on main physical function at twelve months (with some secondary/process differences); and the SupportPrim personalized decision-support trial did not show benefit on primary end points and did worse on one function scale than usual care. Prognostic and machine-learning models (e.g., PICKUP, acute-LBP non-recovery, chronic pain decision trees) typically reach modest discrimination (often around AUC 0.66 or mid-0.6s in validation or cross-validation) with calibration limitations or external validation still required. Digital and AI-augmented tools show a split pattern: selfBACK produced a statistically significant but modest disability difference of uncertain clinical importance; NLP-augmented triage can raise F1 for some referral categories while overall accuracy stays modest and below standalone use; referral-letter screening in other work reaches moderate to good AUC for some conditions in multi-site validation. Case-based reasoning improved device configuration success in one application; SupportPrim similarity retrieval aligned reasonably with prognostic questionnaire distances in development work. DART progressed from excellent usability and pilot triage agreement (weak to moderate raw, higher service-adjusted concordance, no harmful episodes) to a quality-improvement cohort reporting high clinician-tool agreement, operational savings, and high satisfaction—contrasting with null or negative randomized evidence for some other digital decision-support rollouts. DeSSBack pilots suggested medium effects on disability and anxiety and positive clinician views. LLM diagnostic benchmarking (rheumatology) showed

ChatGPT-4 matching or exceeding specialists in some metrics with a sensitivity–specificity trade-off by case type.

ii) Gaps and opportunities

The field shows inconsistent effectiveness of stratified and AI-supported pathways across settings, tools, and trial designs, so generalization from positive early trials is unsafe. Gaps include: definitive trials of tools such as DART and SupportPrim against hard patient-centered outcomes; bridging strong service-level metrics (agreement, throughput) to long-term safety, appropriateness, and equity; prospective external validation of ML prognostic and triage models with decision curve or management consequences; and integrating NLP and LLM components where overall triage accuracy remains modest, including guardrails when specialist-level diagnostic scores mask downstream reasoning or referral appropriateness. Opportunities lie in head-to-head comparisons of stratified care versus digitally augmented pathways, implementation science on fidelity (noted as weak for patients in one CDS pilot), and transparent reporting when pilot success does not match later RCT null results.